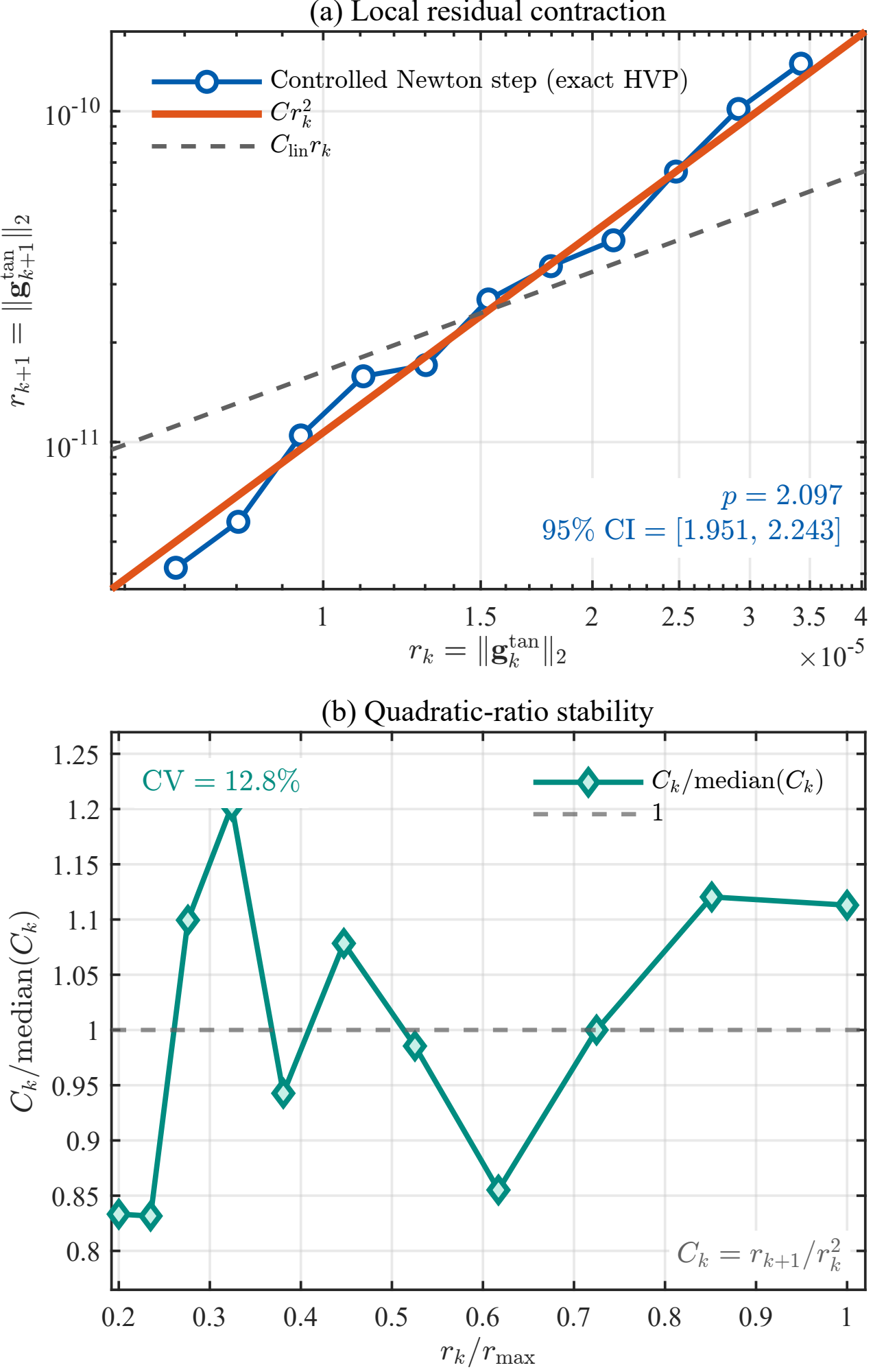

(a) Local residual contraction
Controlled Newton step (exact HVP)
Cr_k^2
$C_{\mathrm{lin}} r_k$
$p = 2.097$
95% CI = [1.951, 2.243]
$r_{k+1} = \|\mathbf{g}_{k+1}^{\tan}\|_2$
$r_k = \|\mathbf{g}_k^{\tan}\|_2$
$\times 10^{-5}$
(b) Quadratic-ratio stability
CV = 12.8%
$C_k/\mathrm{median}(C_k)$
1
$C_k = r_{k+1}/r_k^2$
$C_k/\mathrm{median}(C_k)$
$r_k/r_{\max}$